

Personalized Continuous Blood Pressure Tracking Through Single Channel PPG in Wearable Scenarios

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Abstract—The real-time tracking of human physiopathology states can significantly enhance the quality of personalized healthcare services. Photoplethysmography (PPG) detection is a rapid, portable and non-invasive method for measuring blood flow volume, widely used for monitoring blood pressure (BP) and cardiovascular status. However, continuous BP monitoring technologies based on PPG face numerous challenges in real-world wearable scenarios, such as poor signal quality, complex model computation, and the need for frequent calibration. This work proposed a personalized continuous BP tracking pipeline that performed automatic PPG signal quality grading to reduce the difficulty of model fitting, introduced a lightweight BP model (SCI-GTCN) to alleviate computational complexity, and employed an **adaptive calibration strategy**

to achieve long-term BP monitoring performance under different scenarios. The proposed pipeline was validated using data from 134 subjects in various monitoring scenarios (daytime, nighttime, and abnormal states), assessing the model's performance during rapid BP changes, circadian rhythm fluctuations, and long-term monitoring. The $ME \pm SD$ was $0.99 \pm 7.91/0.36 \pm 5.43$ mmHg. Overall, the results of our method are within the accuracy requirements of the Association for the Advancement of Medical Instrumentation (AAMI) standards, though the subject distribution differs. The method demonstrated good robustness and applicability, making it convenient for deployment on wearable devices and promising in the healthcare field.

Index Terms—Real-world application, single-channel PPG, deep learning, continuous blood pressure, wearable scenarios.

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I. INTRODUCTION

CARDIOVASCULAR disease (CVD) stands as the primary contributor to global mortality [1]. Blood pressure (BP) is a crucial parameter of human body, and it fluctuates under the influence of various factors. Hypertension emerges as one of the risk factors for CVDs, elevating the susceptibility to coronary heart disease, heart failure, atrial fibrillation, and peripheral vascular diseases. It has been demonstrated that the continuous monitoring of BP can effectively discern cardiovascular status, thereby diminishing the occurrence and mortality rates associated with CVDs [2]. The rapid development of intelligent wearable devices and Internet of Things (IoT)-based systems has facilitated the advancement of remote health monitoring. Such applications utilize a set of sensors, communication, and computational infrastructure to continuously track individuals' health conditions, providing real-time monitoring and decision support. Currently, non-invasive continuous BP measurement technology for home daily monitoring has become a research hotspot.

Methods for continuous BP measurement include several approaches. The "gold standard" arterial catheterization method is limited due to its invasiveness. Arterial Tonometry and Volume Clamping methods are relatively mature non-invasive approaches, but they are sensitive to sensor placement and involve complex compensation methods [3]. BP estimation based on pulse transit time (PTT) and pulse wave analysis has gained

attention as a non-invasive method in recent years [4], [5], [6]. However, PTT methods require at least two fully synchronized signals, which are relatively complex. Photoplethysmography (PPG) signals are generated by the pulsations of blood volume changes in the microvascular bed caused by peripheral arterial blood pressure [7]. Due to the physical and geometric properties of the arterial vessels being heterogeneous, parameters such as wall diameter, thickness, and elastic modulus vary along the arterial pathway. This results in changes in the intensity, shape, rhythm, and frequency of the pulse wave as it propagates. These variations carry information related to arterial stiffness, vascular resistance, and blood volume, corresponding to the heart's contraction and relaxation activities as well as dynamic blood pressure fluctuations. Research has shown a high correlation between PPG signal and arterial BP in both the time and frequency domains, making it suitable for BP estimations [4], [8], [9], [10], [11], [12]. And this approach only requires a single PPG sensor, is relatively simple and cost-effective, offering significant advantages in wearable scenarios. However, the application of PPG waveform analysis in real-world scenarios still faces numerous challenges, including: 1) the poor quality of PPG signals due to motion artifacts and noise; 2) the delicate balance between computational complexity and accuracy in the deployment of models on wearable devices; 3) the inter-individual and population variances of PPG signals, which can lead to a degradation in accuracy over long-term periods, and restrict the universality across different scenarios in long-term personalized BP estimates [13].

In conclusion, exploring how to achieve personalized cuffless BP monitoring based on single-channel PPG in wearable scenarios is a significant research direction. It holds great potential to support the application of hypertension management and cardiovascular disease prevention in out-of-hospital settings. This work introduced a continuous BP estimation pipeline designed to track BP fluctuations across multiple scenarios, including abnormal, nighttime and daytime (reflecting abnormal responses, circadian rhythms, and long-term tracking, respectively). The pipeline included automatic PPG signal quality grading, a fusion network based on Sample Convolution and Interaction Network and Gated Time Convolutional Network (SCI-GTCN), and adaptive calibration procedure. The main contributions include:

- 1) Employing an unsupervised clustering approach to automatically grade PPG signals quality, reducing the difficulty of model fitting.
- 2) Introducing a lightweight network (SCI-GTCN), fine-tuned on a pre-trained model to improve model performance.
- 3) Proposing an adaptive calibration strategy to alleviate the challenge of frequent calibration for long-term estimations.
- 4) Constructing a pipeline for dynamic BP estimation based on single channel PPG and deploying it on a wearable system.

II. RELATED WORK

A. PPG Signal Quality Grading

The quality of PPG signals collected by wearable devices can be affected by various factors such as electromagnetic

interference, skin pigmentation, temperature, and movement, which in turn impact the accuracy of blood pressure estimation. It is crucial to ensure signal quality with minimal cost in wearable scenarios. Existing methods for assessing PPG signal quality are based on heuristic rules, statistical features, supervised learning algorithms and unsupervised learning.

Rule-based method applies predefined thresholds on signal quality indices (SQIs), and evaluates whether the PPG signal meets the rules and assigns quality labels [14], [15], [16]. Tiwari et al. [14] explores various SQIs such as skewness and zero-crossing rate, along with a heuristic rule-based system to differentiate PPG data. Rule-based methods exhibit low flexibility and accuracy in complex scenarios.

Statistical feature-based method involves extracting and analyzing statistical features such as mean, variance, and kurtosis from the PPG signal. Chatterjee et al. [17] used statistical methods to evaluate signal-to-noise ratios (SNRs) and waveform shapes to assess signal quality.

Supervised learning-based method uses various machine learning algorithms like K-nearest neighbors (KNN) and support vector machine (SVM) [18] and random forests to classify PPG signals by learning from labeled datasets. Li and Clifford [19] introduced a dynamic time warping method for template matching, using a multilayer perceptron network to classify PPG signal quality. Some research used deep network such as Convolutional Neural Networks (CNNs) [20] and Long Short-Term Memory (LSTM) to extract relevant features automatically from the raw PPG waveform and perform signal quality grading. Supervised learning methods outperform rule-based methods, especially when there is a large amount of training data. However, they are mostly slow and require extensive expert manual annotation of data.

Unsupervised learning-based method achieves automatic PPG grading through clustering and signal segmentation. The common methods include K-means [21], Gaussian Mixture Model (GMM) [22], Fuzzy C-means (FCM) [23] and Autoencoder-Based Clustering (Tripathy et al., 2017; Li et al., 2021). Unsupervised method does not require extensive manual labeling, making it suitable for scenarios with large datasets where manual annotation is challenging, particularly in applications involving wearable devices.

B. Cuff-Less BP Estimation Through Single-Channel PPG

With the development of signal processing and internet technology, some studies have verified the feasibility of continuous BP estimation using a single-channel PPG with machine learning and deep learning algorithms [24], [25], [26] [27], [28], [29], [30]. Zhang et al. [31] extracted 9 features from PPG signals and estimated BP through SVM method, achieving higher accuracy compared to linear regression. Xing et al. [32] combined PPG waveform analysis with biometric techniques and used a random forest algorithm for calibration-free BP estimation across different populations. Some studies have leveraged deep learning algorithms to automatically learn multi-dimensional features. Slapnicar et al. [11] proposed the spectro-temporal ResNet to estimate BP using single-channel PPG waveforms.

Schlesinger et al. [25] introduced a calibration approach using Siamese networks to estimate BP values using only PPG signals. Rong et al. [12] estimated BP by fusing raw PPG signals, PPG morphological, and frequency domain features through a fusion network. Aguirre et al. [26] proposed a sequence-to-sequence LSTM neural network followed by an attention structure, with calibration processing. Harfiya et al. [27] also adopted a sequence-to-sequence model structure and further applied transfer learning methods. Wang et al. [28] created images from PPG segments using a visibility graph, extracted feature vectors through a pre-trained AlexNet, and estimated BP using ridge regression. Roy et al. [30] extracted 20 time-domain, statistical, and auto-encoding features from high-quality PPG signals and estimated BP using a Temporal Convolutional Network (TCN), improving model accuracy. While deep networks can significantly improve the accuracy of BP estimation, they often introduce the challenge of increased computational complexity. A lightweight and accuracy BP estimation model was required for embedded devices.

C. Calibration in Long-Term BP Estimation

Research has shown that the nighttime BP is highly associated with a 5-year risk of cardiovascular death [33]. Existing research mostly focus on the accuracy of short-segment (tens of seconds) BP estimation, lacking validation through long-term experiments. Individual differences in body structures, physiological states, and vascular characteristics may lead to variations in the correlation between BP and PPG signals. Furthermore, the accuracy of the models tends to degrade over time [33], requiring frequent calibration [13]. The long-term effectiveness of BP estimation was improved by setting calibration rules [35], adding error calibration items [25], [33] and introducing individual data into the model [9], [10].

Liu et al. [36] estimated continuous BP values using small artery PTT obtained through multi-wavelength PPG, with all subject-specific variables in their model individually calibrated. Masè et al. [37] validated the stability and accuracy of model parameters for up to 30 days post-calibration. Chen et al. [38] tracked slow variations in SBP from cuff-based BP values, while high-frequency components were fine-tuned from specific frequency bands of PTT to intermittently calibrate BP values. Haddad et al. [39] calibrated BP estimates by applying a simple offset within the same subject. Schlesinger et al. [25] proposed a Siamese network that subtracts the feature vectors output by a pre-trained model from the calibrated feature vectors, updating the activation layers iteratively with the calibrated feature vectors. Song et al. [40] proposed a recalibration system to relearn individual models. Some studies [9], [41] employed transfer learning, which used a small amount of target subject data to fine-tune specific layers of the model to enhance the accuracy of BP estimation.

However, existing calibration strategies still have the problem of reduced reliability of long-term monitoring [33]. There is currently no consensus on the calibration frequency and number of calibrations required for accurate BP monitoring. Mukkamala et al. [13] suggested that the maximum calibration period for BP

monitoring could be at least 6 months, recommending different calibration intervals based on age and gender groups. Choi et al. [42] validated that initial calibration should be based on 40 BP measurements. Additionally, they set calibration intervals of 30, 60, and 90 minutes to assess the impact of calibration frequency on BP estimation accuracy. Yoon et al. [43] suggested that at least four recalibrations, each lasting 60 seconds, were necessary to maintain 12 hours of monitoring, which implies an average recalibration every 3 hours. Longer calibration periods tend to reduce estimation accuracy, while shorter calibration periods, though capable of effectively capturing vascular conditions changes, may cause some disruption to users' daily lives [44].

III. METHOD

The framework of the method is illustrated in Fig. 1. The overall implementation of the pipeline includes sensor data collection, Signal Quality Assessment (SQA), model training and deployment. Firstly, collecting raw PPG and BP data from multiple scenarios. Then, preprocessing the raw PPG signals to grade the signals quality. The SCI-GTCN framework was utilized to construct BP models. Finally, applying adaptive personalized calibration techniques to update the model. The whole pipeline was deployed on a wearable system.

A. Dataset Description

The data utilized were derived from Queensland Database [45], Surgical Dataset, MIMIC III waveform database subset [46], and Multi-day dataset [47]. The combined dataset from the aforementioned four databases contains a wealth of information, with blood pressure tracking times covering short-term (few minutes), medium-term (>12 hours), and long-term (few days), encompassing scenarios of abnormal, nighttime, and daytime. The description of the datasets is provided in Table I.

1) *Abnormal Dataset*: This work conducted an analysis of abnormal BP using data from surgical scenarios, where drug induction during surgery can lead to significant fluctuations in BP (20~80 mmHg in SBP, 10~40 mmHg in DBP). The Surgical Dataset was collected by Tongde Hospital in Zhejiang Province, the experimental procedures were approved by the Clinical Research Ethics Committee of Tongde Hospital (Approval number: 2022-291(K)). All participants gave informed consent prior to inclusion in the study. The data includes continuous arterial blood pressure (ABP) and PPG recordings from 24 patients during various surgical procedures, with an age range of 67 ± 10 years (mean $\pm$ SD) and recording durations ranging from 20 minutes to 7 hours. The Queensland Vital Signs Dataset records data from 32 patients during surgery (25 under general anesthesia, 3 under spinal anesthesia, 4 under sedation) with durations ranging from 13 minutes to 5 hours (median 105 minutes). All cases include ECG, pulse oximeter PLETH, and non-invasive arterial blood pressure data (NIBP), with a few cases incorporating arterial blood pressure (ABP). The sampling rate is 100 Hz.

2) *Nighttime Dataset*: MIMIC-III waveform records include digitized signals such as ECG, ABP, respiration, and PPG, with

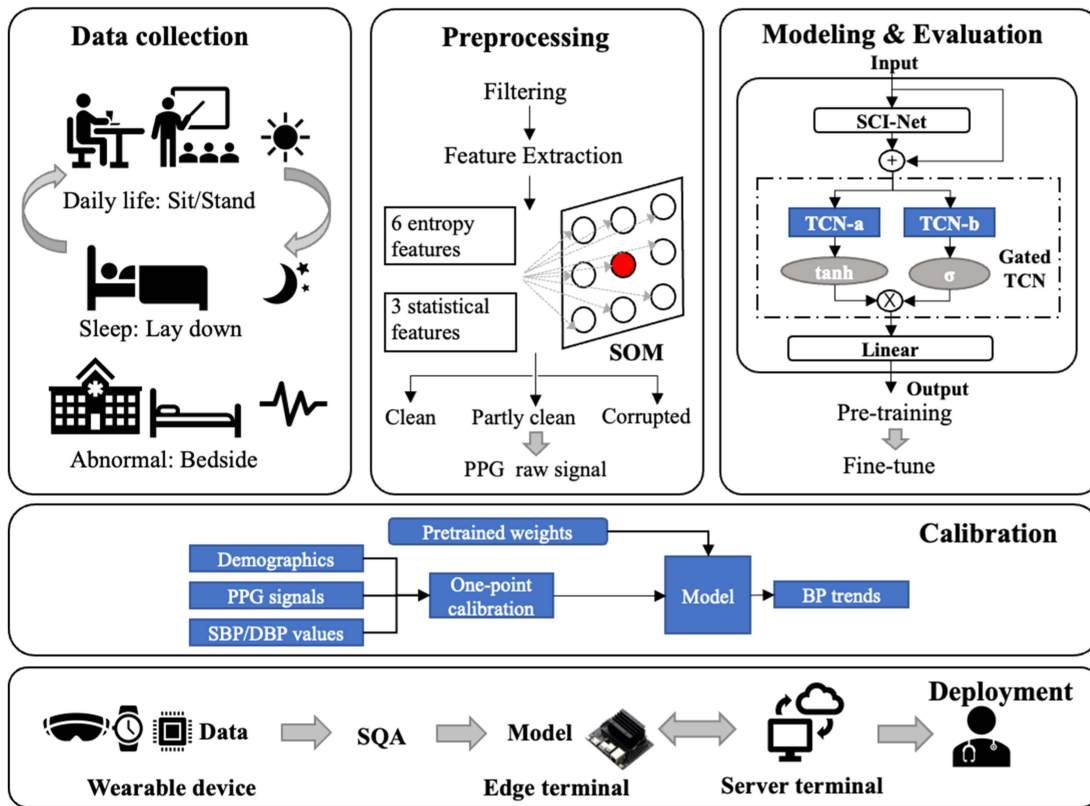


Fig. 1. The framework of the methodology.

TABLE I
DESCRIPTION OF THE DATASETS

Dataset	Method	Scenario	Duration	Subjects
Queensland	Non-invasive, Intermittent	Abnormal	Short-term (13min~5h)	32
Surgical	Invasive, Continuous	Abnormal	Short-term (20min~7h)	24
MIMIC-III	Invasive, Continuous	Nighttime	Medium-term (avg>12h)	66
Multi-day dataset	Non-invasive, Continuous	Daytime	Long-term (6 months)	12

a sampling frequency of 125 Hz. Corresponding clinical records encompass demographics, bedside vital sign measurements, diagnoses, medications, and imaging reports. We selected fingertip PPG and ABP signals recorded during the night, lasting over 8 hours to validate the model's performance in nighttime scenarios, which include the information about the circadian pattern of BP. Additionally, we selected data from normal populations (33 subjects) and CVD populations (33 subjects with various typical CVDs, such as rheumatic heart disease, acute myocardial infarction, angina, coronary atherosclerosis, atrial fibrillation, etc.), with an average length of 15.48 hours. Among them, there were 36 males and 30 females, with an age distribution of 56.3 ± 19 (mean $\pm$ SD).

3) Daytime Dataset: Long-term BP tracking (spanning months) data is challenging to obtain, and few studies evaluate BP over such extended periods. To observe the long-term patterns of individual BP fluctuations, this work conducted a multi-day longitudinal experiment on 12 healthy individuals, aimed at preliminarily assessing the model's long-term performance in daytime settings. The Multiday experiment was approved by

Joint Chinese University of Hong Kong-New Territories East Cluster Clinical Research Ethics Committee (Approval number: 2017-484), and all subjects gave their informed consent prior to the experiment. The dataset including ECG and PPG signals acquired from Biopac system, and reference continuous BP measured by Finapres system simultaneously. The BP, ECG and PPG data of each subject were recorded at sampling frequency of 1000 Hz for 8 minutes at the rest status in a multi-day period, namely 1st day, 2nd day, 4th day and 6 months after the first day.

B. PPG Automatic Signal Quality Grading

We divided each record into segments of 10 seconds and extracted six entropy-based features (approximate entropy, sample entropy, Shannon entropy, spectral entropy, permutation entropy, multiscale entropy) and three statistical features (skewness, kurtosis, Higuchi fractal dimension) from each PPG segment for signal quality assessment, as detailed in Table II.

To reduce the workload of manual annotation and enhance the automation of the algorithm, we investigated an unsupervised

TABLE II
FEATURES FOR PPG SIGNAL QUALITY ANALYSIS

Features	Details
Skewness	Asymmetry
Kurtosis	Flatness
Approx_entropy	Complexity
Samp_entropy	Complexity
Shan_entropy	Uncertainty
Spec_entropy	Complexity
Permutation_entropy	Periodic randomness
Multi_scale_entropy	Complexity at different time scales
Higuchi Fractal Dimension	Nonlinear measure of complexity

clustering approach using the Self Organizing Maps (SOM) algorithm to grade the signal quality of PPG segments. The SOM algorithm assumes the existence of some topology or order in the input objects, representing points in a high-dimensional space with points in a low-dimensional target space, while preserving the distances and neighborhood relationships (topological relationships) of corresponding points. This achieves a dimensionality reduction mapping from the input space (n-dimensional) to the output plane (2-dimensional).

Nine features were extracted from each 10-second PPG segment as input nodes, and a 5*5 fully connected SOM was employed to classify signal quality, identifying segments as “clean,” “partially clean,” or “corrupted.” For each output node k , the connection weights between the input layer and the output layer were represented as $W_k = (w_{1k}, w_{2k}, \dots, w_{nk})^T$. Weight adjustments were made based on the Euclidean distance D_t for each output unit. The output unit with the shortest distance, termed the winning unit, is the closest to a specific input sample.

$$D_t = \sqrt{\sum_{j=1}^n (I_j - W_{ij})^2} \quad (1)$$

During the testing phase, PPG feature vectors were input into the SOM, and the winning neuron was identified. The Euclidean distance between the feature vector and the winning neuron was calculated, and input PPG windows belonging to the same class will cluster in the output layer. Signal quality was classified into three levels during testing, excluding segments with signal damage, and the remaining signals are used for model construction.

In this work, we meticulously identified 2041 quintessential PPG segments for expert annotation, employing this to substantiate our signal quality grading methodology. These segments were systematically categorized into ‘clean,’ ‘moderately clean,’ and ‘corrupted’ classes, and were assigned the numerical labels 0, 1, and 2, respectively, to facilitate the evaluation of the grading system’s accuracy and reliability. Based on relevant studies on PPG signal quality assessment [20], [48], [49], we have established the rules to classify PPG signals into three categories, the quality levels were defined by the percentage of the segment for which clear pulses with discernible peaks were identifiable:

- Class 0: Clean and complete signal, with recognizable PPG pulse in $\geq 75\%$ of a 10-second segment.

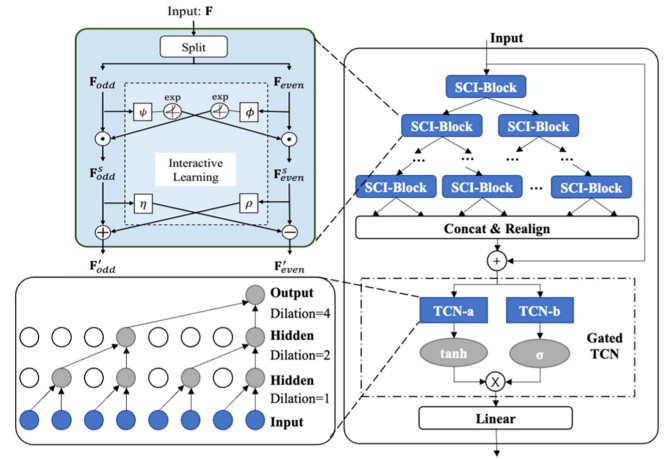


Fig. 2. The structure of SCI-GTCN model.

- Class 1: Partially complete signal, with recognizable PPG pulse in $\geq 50\%$ of a 10-second segment.
- Class 2: No waveform or segments of noise, with recognizable PPG pulse in $< 50\%$ of a 10-second segment.

C. Estimation Model

Blood pressure fluctuates with physiological cycles, circadian rhythms, and lifestyle habits. And due to individual variations in vascular properties, the patterns of BP change are quite complex. In wearable device scenarios, the computational resources of hardware are typically limited, making it essential to use models that are highly parallelizable and optimized for efficiency. Convolutional networks (CNNs) are well-suited for this computational paradigm, but modeling convolutional sequences requires a broader consideration of the correlations between sequence processing and recurrent networks. This work proposed a novel architecture (SCI-GTCN) that integrates various network modules (as Fig. 2). The network employs Sample Convolution and Interaction Network (SCINet) as a feature extractor to obtain multi-temporal granularity information from raw PPG sequences. The tanh-gated temporal convolutional network is employed to learn intricate temporal dependencies.

SCINet adopts an encoder-decoder architecture, with the encoder being a hierarchical convolutional network capturing dynamic temporal dependencies at multiple resolutions [50]. The SCI-Block subsamples the input data or features into two subsequences, processes each subsequence with a set of convolutional filters, extracting distinct yet valuable time features from each part. To compensate for information loss during subsampling, interactive learning is applied between the two subsequences. SCINet constructs a binary tree structure by arranging multiple SCI-Blocks, each providing a local and global view of the entire time sequence. After all subsampling-convolution-interaction operations, the extracted features are rearranged into a new sequence representation and added to the original time sequence, with a fully connected network serving as the decoder for prediction. The SCINet can capture multi-scale patterns, effectively learning different temporal resolutions to account for the

periodic and non-periodic variations in BP. Multi-temporal granularity data supplements single-granularity features, where coarse granularity reflects overall trend changes, and fine granularity represents subtle waveform variations. From a mathematical perspective, SCINet extracts patterns in both frequency and time domains, allowing the model to handle the inherent complexity of physiological signals, where useful information may be buried in different frequency bands.

TCN is constructed by stacking causal convolutional layers with an exponential expansion factor, achieving a large receptive field with a small number of convolutional layers. The convolutions in the TCN architecture are causal, which can handle sequences of arbitrary length, mapping them to an output sequence of the same length. Additionally, TCN uses a deep network (enhanced with residual layers) and a combination of dilated convolutions to build a very large effective history size. For blood pressure estimation, TCN mathematically models long-term dependencies in PPG sequence data through dilated convolutions, helping to capture short- and long-term variations in PPG signals. By combining residual connections and non-linear activation functions, TCNs stabilize the learning process and enable the network to adapt to diverse PPG signal dynamics. TCN has found widespread applications in various time series prediction problems, demonstrating higher accuracy and simplicity compared to RNNs and LSTM networks [51].

Gate mechanisms control the flow of information based on specific PPG features by performing element-wise multiplications with learned weights, similarly to the gating units in Long Short-Term Memory (LSTM) networks or Gated Recurrent Units (GRUs). This design helps the network handle sequential data more effectively by selectively passing relevant information and discarding less relevant information, improving both accuracy and efficiency. Gated Linear Units reduce the gradient vanishing problem in deep architectures by providing a linear path for gradients while preserving nonlinear functionality. This is especially valuable when working with long sequences, as it allows gradients to propagate more effectively through the network.

In summary, the SCI-GTCN model leverages PPG signals and applies mathematical transformations that enhance its ability to capture subtle, underlying patterns related to BP. The model utilizes a convolutional structure instead of recursive connections in recurrent networks, resulting in a lower parameter count and computational complexity (142k parameters, lower than models in other studies [9]).

D. Calibration

Recalibration has been identified as a necessary procedure to ensure accurate estimation of cuffless BP monitoring systems, and it is recommended to use cuff-based monitors for regular calibration.

This work proposed a generalizable and adaptive calibration strategy, as shown in Fig. 3, with the potential to achieve stable and accurate long-term individual BP monitoring in real-world applications. The proposed calibration strategy assumes two data modes or scenarios in the deep learning model: 1) Individual

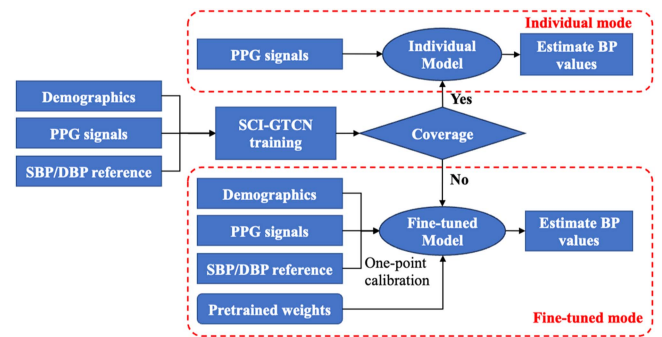


Fig. 3. Personalized calibration strategy.

Mode: When data volume is sufficient for the model to converge, all network parameters are updated to create an individual model. The model's convergence point was defined as the moment when the loss stabilizes and stops decreasing for 8 consecutive epochs. 2) Fine-tuned Mode: When data is insufficient to converge, only certain network parameters are updated based on a pre-trained model, creating a fine-tuned model. The adaptive calibration strategy evaluates model loss continuously and switches from fine-tuned to individual mode once each user's model reaches its optimal convergence point, personalizing the model.

In real-world wearable scenarios, a pre-trained SCI-GTCN model was initially developed using large-scale, heterogeneous sample data from our previous research [16], providing initial weights for the model. A new user gathers a small amount of personal PPG and BP data using a wearable device, fine-tuning certain network layer parameters through transfer learning based on the pre-trained model. This process allows for model updating and BP calculation. Optionally, the model can be further calibrated by incorporating demographic information such as age, gender, medical history, and BMI to account for individual variations. Subsequent periodic recalibration occurs by collecting intermittent PPG and reference BP values to adjust model parameters: short- and medium-term data is updated every 30 minutes, while long-term data is updated once on the first day. Once enough data has been collected for the model to reach convergence, all parameters are updated to build an individualized model. And then BP can be tracked continuously without further calibration under consistent physiological conditions.

IV. RESULTS

A. Experimental Setup

The model was implemented using PyTorch on an Intel i5 2.9 GHz computer with 32 GB RAM. NVIDIA GeForce 3090 GPU was utilized for deep network training. The proposed model was evaluated using Mean Absolute Error (MAE) as a metric, and the Mean Error (ME) and Standard Deviation (SD) of the errors were computed.

B. PPG Automatic Grading Results

A total of 2041 typical PPG segments were selected for expert annotation, categorized as clean, partly clean, and corrupted

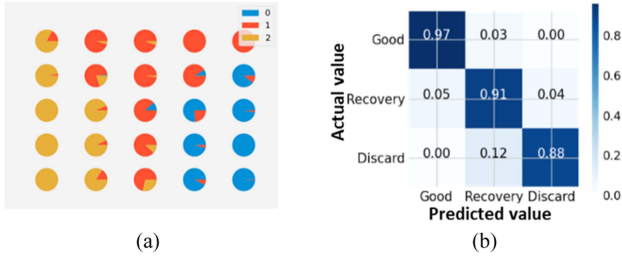


Fig. 4. Grouping of three classes in SOM output layer.

TABLE III
RESULTS OF PPG SIGNAL QUALITY CLASSIFICATION

Class	Precision	Recall	F1-score
Clean	0.94	0.97	0.95
Partly clean	0.90	0.91	0.90
Corrupted	0.92	0.88	0.90

TABLE IV
RESULTS OF THE SCI-GTCN MODEL IN VARIOUS SCENARIOS

	Dataset	MAE	ME	SD
SBP (mmHg)	Abnormal	9.66	2.97	9.57
	Nighttime	9.93	-0.03	9.38
	Daytime	7.08	0.02	4.78
	All	8.89	0.99	7.91
DBP (mmHg)	Abnormal	5.82	0.73	5.87
	Nighttime	6.14	0.59	6.14
	Daytime	5.53	-0.25	4.27
	All	5.83	0.36	5.43
AAMI			≤ 5	≤ 8

(label as 0, 1 and 2) in a ratio of 677:874:490. Nine features extracted from each PPG signal were fed into a SOM classifier. The distribution of the three classes in the SOM output layer is illustrated in Fig. 4. Some classes are associated with different sections of the SOM output layer, while others concentrate in a limited region. The performance of SOM for multi-class classification of PPG segments is presented in Table III, with average accuracy, recall, and F1 score values of 91.72%, 91.94%, and 91.55%, respectively.

C. SCI-GTCN Individual Model for BP Estimation

We constructed individual models for each subject from three datasets and selected the size of training data based on the model's convergence results, aiming to achieve model performance with the smallest training quantity. The final amount of data used for model training accounted for approximately 19% of the total data. For abnormal datasets, the average training data was 152 segments (each segment lasting 10 seconds), approximately 25.33 minutes; for nighttime datasets, it was 715 segments, approximately 1.99 hours; and for daytime datasets, it was 48 segments, about 8 minutes. Table IV records the errors between estimated and true BP values. The comparison of BP estimates and true values recorded by three typical subjects is shown in Fig. 5, with the x-axis representing durations and the y-axis indicating BP values.

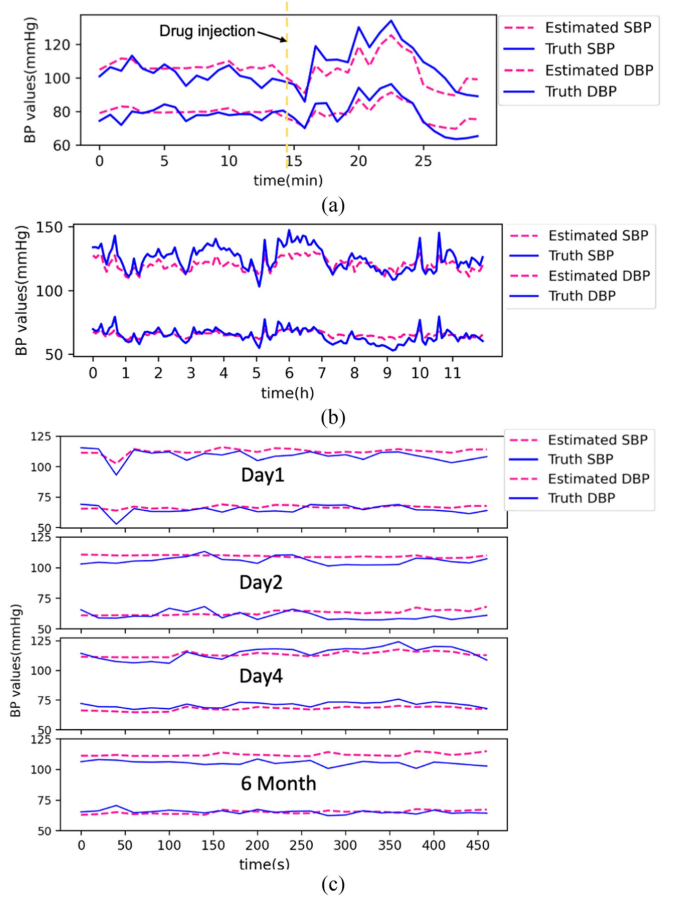


Fig. 5. Results comparison on specific subjects in (a) abnormal dataset (ID: Ab18) (b) nighttime dataset (ID: Ni10) (c) daytime dataset (ID: Da8).

1) *Short-Term Tracking of Abnormal Blood Pressure*: Evaluated through a total of 56 records from 32 subjects in the Queensland dataset and Surgical dataset. The MAE values of SBP/DBP are 9.66 mmHg/5.82 mmHg. This may be due to the significant BP fluctuations induced by drug, as the performance of BP estimation models based on single PPG signals is limited when facing a wide dynamic range. A typical record (ID: Ab18) from the Surgical dataset is shown in Fig. 5(a), where the dashed line indicates the starting point of drug (Metaraminol) injection. The drug-induced effect leads to a sharp increase in BP, and the model can effectively track its fluctuations.

2) *Mid-Term Tracking of Nighttime Blood Pressure*: Evaluated through 66 nighttime records from the MIMIC dataset. The MAE values of SBP/DBP were 9.93 mmHg/6.14 mmHg. Fig. 5(b) presents a typical record (ID: Ni10) result, indicating that despite the challenges in accurately estimating the absolute BP values, the model effectively tracked the fluctuations during nighttime, capturing both the increases and decreases in BP.

3) *Long-Term Tracking of Daytime Blood Pressure*: Evaluated through 12 subjects from the Multi-day dataset. The model was with part of the 1st day data and tested on the rest of the 1st day data as well as the following days' data. The MAE for SBP/DBP estimation was 7.08 mmHg/5.53 mmHg, as shown

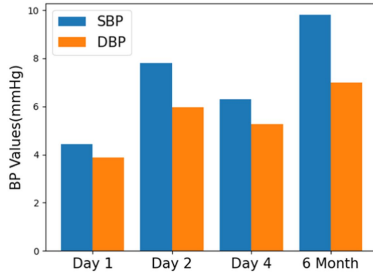


Fig. 6. The MAE results on daytime dataset.

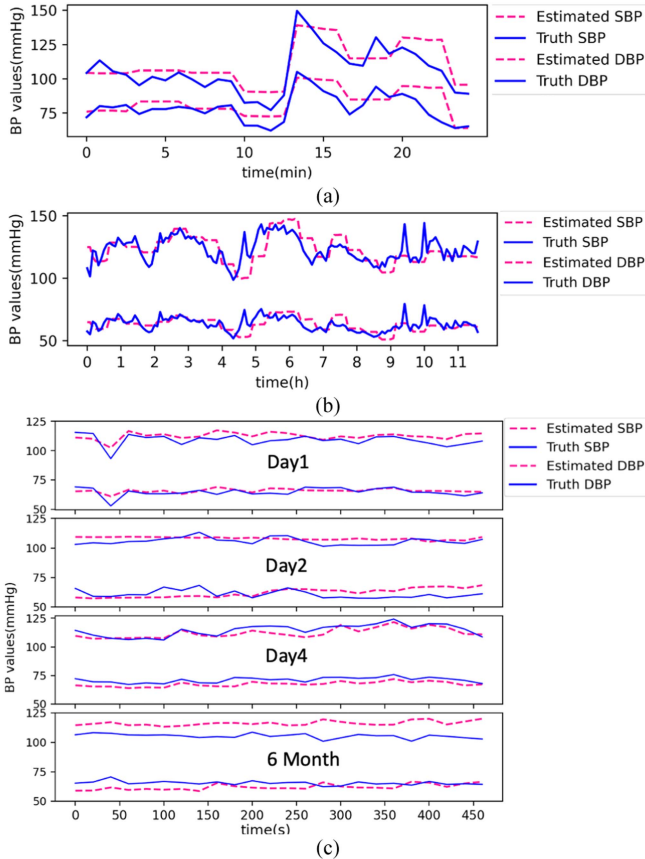


Fig. 7. Fine-tuned model's performance on specific subjects in (a) abnormal dataset (ID: Ab18) (b) nighttime dataset (ID: Ni10) (c) daytime dataset (ID: Da8).

in Table IV. And for multi-days period, it achieves 4.44, 7.81, 6.30 and 9.80 mmHg for the 1st day, 2nd day, 4th day and 6th month after the 1st day SBP estimation, and 3.87, 5.97, 5.27 and 6.99 mmHg for DBP estimation, respectively. Fig. 5(c) displays the BP tracking results for a specific individual (ID: Da8) over multi-days, the four subgraphs correspond to the results of 1st day, 2nd day, 4th day and 6th month respectively. Fig. 6 shows the MAE between estimated and true values on the whole dataset. The model can track the long-term trend of individuals over 6 months with a small error, but its error increases over time, consistent with the conclusions in related studies [47], [52]. The relatively low MAE observed on day 4 can be attributed to limitations in the current dataset. As the dataset consists of short

TABLE V
ERROR ANALYSIS OF FINE-TUNE MODEL/INDIVIDUAL MODEL FOR THREE SPECIFIC SUBJECTS (ID: Ab18, Ni10, Da8)

record ID	Model	SBP (mmHg)			DBP (mmHg)		
		MAE	ME	SD	MAE	ME	SD
Ab18	Fine-Tune	9.25	5.05	10.14	7.05	4.18	8.20
	Individual	7.64	-0.42	9.47	4.88	0.87	6.21
Ni10	Fine-Tune	7.90	0.03	10.73	4.54	-0.27	6.34
	Individual	7.07	-4.88	7.20	3.50	0.73	4.70
Da8	Fine-Tune	6.31	4.47	4.32	4.19	-1.21	4.19
	Individual	5.61	4.15	4.22	3.54	0.21	3.68

8-minute segments under controlled conditions, with relatively stable BP values and minimal fluctuation. For such limited and narrow-range data, the robustness of the trained model may not be optimal. Additionally, the small sample size introduced a degree of randomness in the results.

This section evaluated the effectiveness of BP estimation under various scenarios using the above-mentioned multi-scenario datasets and compared the results with the Association for the Advancement of Medical Instrumentation (AAMI) standards. The dataset includes 134 participants, exceeding the AAMI's minimum requirement of 85 participants. Moreover, our work involved several hours of continuous beat-to-beat BP measurements, far surpassing the AAMI's minimum requirement of three readings per participant. Despite these heightened challenges, the $ME \pm SD$ for SBP and DBP are 0.99 ± 7.91 mmHg and 0.36 ± 5.43 mmHg, respectively, as shown in Table IV, both within the accuracy requirements of the AAMI standard ($ME \pm SD \leq 5 \pm 8$ mmHg)[53].

D. SCI-GTCN Fine-Tuned Model for BP Estimation

To evaluate the effectiveness of the calibration strategy, fine-tuned model was verified. Three typical subject records from the three scenarios were selected for evaluation (consistent with the subjects in Fig. 5), with short-term data lengths of 38.3 min, mid-term data lengths of 15.6 h and long-term data with multi-days. The parameter weights of the first few layers were fixed, and only the last layer was updated to estimate BP values. The model was intermittently fine-tuned with a small amount of individual data. The estimated errors (MAE, ME, SD) of SBP/DBP are presented in Table V.

For the short-term abnormal record (ID: Ab18), the $ME \pm SD$ for SBP/DBP estimation was 0.32 ± 6.93 mmHg/ 0.82 ± 5.96 mmHg; for the mid-term nighttime record (ID: Ni10), the error was 0.03 ± 10.73 mmHg/ -0.27 ± 6.34 mmHg; and for the long-term daytime record (ID: Da8), the error was 4.47 ± 4.32 mmHg/ -1.21 ± 4.19 mmHg. The SBP/DBP estimates versus true values for three typical subject records are compared in the Fig. 7, with the x-axis representing the duration. After intermittent calibration, the fine-tuned model accurately tracked BP fluctuations over an extended period in different populations.

Here, we compared the error results of individual model and fine-tuned model for typical subjects, as shown in Table V. For

TABLE VI

RESULTS OF FEATURE ABLATION IN PPG SIGNAL QUALITY CLASSIFICATION

Features	Precision	Recall	F1-score
w/o entropy features	0.87	0.86	0.87
w/o statistical features	0.89	0.88	0.89
All features	0.92	0.92	0.92

the short-term abnormal record, the $ME \pm SD$ for SBP/DBP estimation was 0.84 ± 3.15 mmHg / 1.34 ± 3.44 mmHg; for the mid-term nighttime record was -4.88 ± 7.19 mmHg / 0.73 ± 4.70 mmHg; and for the long-term daytime record was 4.15 ± 4.22 mmHg / 0.21 ± 3.68 mmHg. The SBP and DBP estimation errors for typical subjects in all three scenarios had lower MAE compared to the fine-tuned model, indicating that individual models are more suitable for personalized continuous BP estimation.

It can be concluded that our proposed calibration strategy exhibited good robustness and applicability, accurately tracking continuous BP values across diverse populations in different scenarios.

E. Ablation Study

We conducted an ablation study focusing on feature selection for signal quality classification in the preprocessing phase, as well as on model architecture in the modeling phase.

1) *Ablation of Feature Selection*: Ablation study with various feature combinations was conducted to evaluate the impact of the statistical and entropy features selected in this work on PPG signal quality grading. The input features include entropy features and statistical features. Entropy features quantify irregularity and unpredictability, aiding in distinguishing whether abnormalities in PPG signals arise from physiological complexity or data corruption. Statistical features assess the shape and distribution of the signal. While entropy captures the overall variability and irregularity of a signal over time, helping to evaluate global consistency, statistical features focus on the symmetry and morphology of individual waveforms. The experimental setup was identical to that in Section B. We evaluated the grading results under different feature combinations using precision, recall, and F1 score metrics, as shown in Table VI. The results indicate that the results incorporating all features outperformed those without entropy features and those without statistical features, showing improvements in accuracy of 5.75% and 3.37%, respectively.

2) *Comparisons of PPG Signal Quality Grading Method*: As the main goal of our work is to estimate BP values in wearable scenarios, so we used a simple and unsupervised clustering network (SOM), which requires no labeling or model training for automated PPG signal quality grading. For consistency and fairness, we compared with other similar automated PPG signal grading methods, such as rule-based, K-nearest neighbor, Gaussian Mixture Model and fuzzy C-means clustering. The experimental setup was identical to that in Section B, using all 9 features as input, while replacing SOM with different grading methods for comparison. Referring to signal selection rules from

TABLE VII

COMPARISON OF PPG SIGNAL QUALITY CLASSIFICATION RESULTS BASED ON DIFFERENT METHODS

Method	Precision	Recall	F1-score
Rule-based [16]	0.70	0.68	0.68
Kmeans [18]	0.74	0.73	0.74
GMM [22]	0.75	0.74	0.75
FCM [23]	0.84	0.86	0.85
SOM	0.92	0.92	0.92

TABLE VIII

RESULTS OF MODEL ABLATION IN BP ESTIMATION

Model	SBP (mmHg)		DBP (mmHg)	
	RMSE	MAE	RMSE	MAE
TCN	9.55	7.66	7.50	6.00
GTCN	9.36	7.47	7.45	5.93
SCI-TCN	9.40	7.53	7.42	5.91
SCI-GTCN	9.28	7.44	7.20	5.70

previous work [16], we proposed a process to categorize signals into three classes. As shown in Table VII, rule-based methods have lower accuracy due to their lower flexibility, making them unsuitable for dynamic signal changes. In comparison, the FCM algorithm shows improvement in accuracy, but our method demonstrates superior performance in terms of classification accuracy, recall rate, and F1-scores. This is because the SOM algorithm can automatically find intrinsic patterns and essential attributes in the samples, self-organize, and adaptively change network parameters and structures, exhibiting better flexibility and performance.

3) *Ablation of SCI-GTCN Architecture*: We conducted an ablation study on the entire SCI-GTCN architecture to evaluate the impact of each module. The experimental setup was identical to that in Section C, using root mean square error (RMSE) and mean absolute error (MAE) as metrics. The study randomly selected 25 participant records from the nighttime dataset for comparative modeling, resulting in a total of 74192 segments. The nighttime dataset is a subset of the MIMIC-III database, chosen for its long continuous BP recordings (over 12 hours), which captures various blood pressure fluctuations. As MIMIC is the most extensively studied open database, it is well-supported by related research. First, we assessed the baseline TCN model's performance. By incorporating a gating mechanism to control information flow, the GTCN model achieved improved performance over the baseline. Adding SCINet to capture multi-temporal PPG sequences further enhanced model accuracy. Finally, the results were compared with the baseline model, demonstrating that the TCN, SCINet, and gated mechanism modules all contribute to the overall model performance, as shown in Table VIII. The SBP/DBP estimation accuracy improved by 2.4%/1.2% with GTCN, 1.7%/1.5% with SCI-TCN, and 2.9%/5% with SCI-GTCN.

V. DISCUSSION

A. Comparison of Model's Performance

Currently, most research on estimating blood pressure through photoplethysmography (PPG) is limited to short-term blood

TABLE IX
COMPARISON OF THE RELATED WORK

	Signal	Model	Subjects	Time length	SBP (mmHg)		DBP (mmHg)		Calibration
					MAE	ME $\pm$ SD	MAE	ME $\pm$ SD	
Leitner [9]	PPG	CRNN	50	1 h	3.5	0.1 $\pm$ 4.6	2.2	0.1 $\pm$ 2.8	Single
Landry [54]	ECG、PPG	NARX	5	4.67 h		-0.8 $\pm$ 8.7		0.2 $\pm$ 7.1	Single
Landry [55]	ECG、PPG	ANN	15	54~320 min		1.2 $\pm$ 17.8		0.6 $\pm$ 8.4	No
Cao [56]	Multi- channel PPG	BiLSTM	35	24 h		0.4 $\pm$ 12.0 1.7 $\pm$ 7.3		0.5 $\pm$ 5.8 0.9 $\pm$ 6.6	Multi No
Qiu [57]	ECG、PPG	Pattern-fusion	85	<10 min >20 h >1 month	3.7 6.8 6.2	0.07 $\pm$ 4.9 0.2 $\pm$ 8.8 0.3 $\pm$ 8.0	4.6 3.8 3.7	1.4 $\pm$ 5.6 -0.1 $\pm$ 5.0 0.2 $\pm$ 4.7	Single
Liu [58]	PPG	BiGRU-Attention	15	average 14.16 h		-0.8 $\pm$ 9.9		0.8 $\pm$ 4.9	No
This work	PPG	SCI-GTCN	134	>12 h, 6 months	8.9	1.0 $\pm$ 7.9	5.8	0.4 $\pm$ 5.4	No

pressure level estimation or intermittent assessments based on regular follow-ups. Evaluations for continuous BP estimation over longer time intervals or in different scenarios are scarce, and there is also a lack of assessment regarding calibration effectiveness. On the other hand, most data-driven methods used a large amount of training data to evaluate the computational performance of the model on short segments, which may lead to issues of data leakage or model overfitting. To further evaluate the model's effectiveness in estimating continuous BP values, we compared it with related work covering longer time ranges, as shown in Table IX.

Leitner [9] applied transfer learning, fine-tuning specific layers of the model with template data from a small number of target subjects, achieving good results, but with limited data length for evaluation. Landry et al. [54] estimated BP waveform using both ECG and PPG signals, evaluating the model's effectiveness in daily life activity scenarios. The number of subjects was limited, and time length was short. Furthermore, they selected 15 subjects from the MIMIC database and evaluated the model under both prediction and intermittent training modes [55]. While their work made progress in BP estimation, there is room for improvement in error results. Additionally, obtaining ECG and PPG signals simultaneously adds complexity to practical applications. Cao et al. [56] monitored continuous BP using wearable devices with multi-wavelength PPG, meeting AAMI standards. However, their subjects were all from healthy population, and the experimental setup involved short-term assessments every 0.5/1 hour, not strict continuous monitoring. Qiu et al. [57] employed a model fusion approach, incorporating cardiovascular coupling effects into the vascular model, achieving good results in different scenarios, but their results heavily depended on the calibration process and required obtaining two signals simultaneously. Liu et al. [58] explored the estimation of individual BP over a long period of time in natural scenarios, however, their work involved a limited number of subjects and achieved suboptimal results. This work evaluated the performance of the model on data from 134 subjects across various scenarios, using only a limited amount of training data. Additionally, calibration strategies were established to offer options for both calibration-free and calibration modes based on the volume of data available. The model estimated continuous

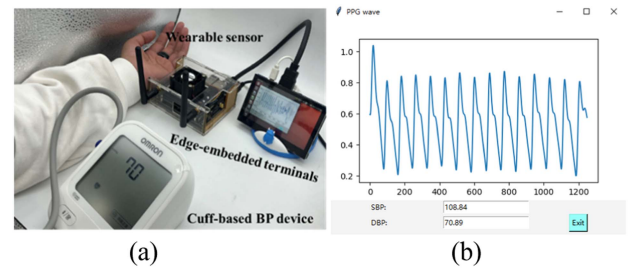


Fig. 8. (a) The deployment of the pipeline, (b) the GUI display.

SBP/DBP values through single-channel PPG signals, demonstrating robustness and universality.

B. Pipeline Deployment

This section introduces the initial physical implementation of our proposed method and optimization strategies (as illustrated in the Fig. 8), which includes the setup of wearable sensors, cuff-based BP calibration devices, embedded terminals, and a cloud server environment, integrating the BP estimation model into practical application scenarios. The process involves the following steps: establishing communication links between wearable sensor and the embedded terminal, configuring the terminal environment, exporting and deploying the SCI-GTCN model, and performing BP inference computations. Additionally, a GUI graphical interface can be invoked on the embedded terminal, enabling functions such as user information input, model calculation, and result display.

The real-time performance of our method was evaluated, including model training and inference times on the Jetson nano terminal using its onboard GPU and TensorRT acceleration. For testing, we randomly selected one record, using a PPG segment updated every 10 seconds to output SBP and DBP values. Without model compression or optimization at this stage, real-time performance is not fully optimized, but it meets the set update frequency. Initial model fine-tuning takes about 10.8 seconds, while inference time is around 1.92 seconds. Compared to similar works [30], this represents a performance improvement of 23.2%.

C. Limitations and Future Directions

This work provided a pipeline for developing the method to evaluate low-cost, accurate, and real-time BP estimation algorithms and presented some practical applications.

However, there are several limitations in this work: 1) Our work graded the quality of PPG signals and discarded corrupted ones, which led to a reduction in the amount of valid data; 2) the proposed deep network still has a high computational demand, resulting in suboptimal performance on wearable platforms; 3) the BP model based on single-channel PPG has limited performance and cannot fully reflect the human BP regulation process, showing inferior results in large heterogeneous populations and long-term follow-ups; and 4) the proposed calibration strategy is only a preliminary concept, and a robust calibration method still needs to be explored.

To address those limitations in the future study, we proposed to: 1) consider reconstructing corrupted PPG signals to maximize the use of the original signals; 2) compress and optimize the deep learning models to reduce inference time on edge devices and enhance computational efficiency; 3) utilize multimodal signals for modeling to encompass more cardiovascular information and individual-specific data; and 4) further investigate the number of calibrations, the amount of calibration data, and the calibration cycles required for accurate BP monitoring.

VI. CONCLUSION

Continuous BP estimation through PPG in wearable scenarios faces numerous challenges. This work proposed a systematic pipeline, including signal grading, lightweight model development, and a personalized calibration strategy. Firstly, we performed automated, unsupervised screening of raw PPG signals to reduce model fitting complexity and enhance robustness. Secondly, we proposed a lightweight network, SCIGTCN, which extracted multi-granularity temporal information and long-range dependency from raw PPG signals. The model exhibited stronger modeling and learning capabilities, efficiently performing BP estimation under limited resource. Besides, we proposed a personalized calibration strategy, adapting calibration to individuals in different scenarios to ensure the accuracy and stability of the BP model. Finally, the pipeline was deployed to wearable embedded devices. To our knowledge, there was limited research focusing on continuous BP tracking over extended periods—exceeding 12 hours or even spinning several months. Validation with data from diverse scenarios and timescales demonstrates significant improvements over similar work, indicating its capability to achieve continuous, high-precision dynamic BP tracking in wearable scenarios, potentially contributing to the field of mobile healthcare.

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